

UDAY RAJ

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LinkedIn

GitHub

Portfolio

Education

Indian Institute Of Information Technology, Design and Manufacturing

Kurnool

Bachelor of Technology in Artificial Intelligence and Data Science - **Grade : 8.38**

2024 - 2028

Delhi Public School Sushant Lok

Gurugram

High School - (Physics, Chemistry, Mathematics) - **Grade : 95%**

2022 - 2024

Relevant Coursework

- Data Structures
- Algorithms Analysis
- Computer Vision
- Artificial Intelligence
- Machine Learning
- Deep Learning
- Statistical Analysis
- Discrete Mathematics

Experience

SVNIT Surat

Nov 2025 - Apr 2026

Research Summer Intern - Department of Artificial Intelligence

Surat, India

- Designed TransEvade under Dr. Praveen Kumar Chandaliya, a printability-constrained attention-hijacking attack for transformer-based Person Re-ID systems.
- Engineered a SegFormer/MiT Re-ID pipeline across Market-1501, DukeMTMC-reID, and MSMT17 using MiT-B0/B1 backbones, 256/512-D descriptors, and cosine retrieval.
- Optimized 500-1000 step adversarial pattern generation with Adam, ImageNet normalization, chest/waist/legs masks, and distract-suppress attention losses.
- Quantified 90-identity spatial ablations where torso patches reduced adversarial Rank-1 to 1.67%-7.04% on Duke/MSMT, exposing attention priors.

Projects

CourtSense AI - 3D Tennis Analytics Engine

[GitHub]

- Created an end-to-end sports CV pipeline converting tennis videos into 150-frame 3D replay clips with FastAPI, React, Three.js, and JSON coordinate streams.
- Combined YOLOv8-Pose, SegFormer-B2, OpenCV homography, and HSV/YOLO ball tracking to map 17 COCO keypoints onto court coordinates.
- Applied Kalman filtering, EMA smoothing ($\alpha = 0.3$), 5-frame rolling stats, Magnus-effect correction, km/h speed, and rpm spin estimation.

PaliGemma Vision-Language Model From Scratch

[GitHub]

- Recreated a PaliGemma-style VLM in PyTorch by pairing a 12-layer SigLIP vision transformer, 16x16 patch embeddings, 196 visual tokens, and Gemma decoder.
- Programmed a 768-to-2048 multimodal projector, image-token merging at index 256000, RMSNorm, RoPE, grouped KV attention, and 8192-token context support.
- Enabled KV-cache and causal masking for autoregressive multimodal generation across modular SigLIP, Gemma, and PaliGemma components.

Technical Skills

Languages: Python, C, C++, SQL, HTML

Machine Learning Stack: PyTorch, TensorFlow, Keras, torchvision, OpenCV, scikit-learn

Computer Vision & Deep Learning: YOLOv8, SegFormer, ViT, SigLIP, Person Re-ID, Transformers, NumPy, Pandas

MLOps & Deployment: FastAPI, REST API, Docker, CUDA, Linux, Git, GitHub, Jupyter

Achievements

- Placed global top 20 in the [Shell.ai] Hackathon among 1000+ submissions, optimizing ML model parameters.
- Ranked 176th out of 7000+ registrants in [AWS x Zelestra ML Ascend Hackathon].
- Earned a 3-star rating of 1630 on [CodeChef] and solved 100+ DSA problems on [LeetCode] and [Code360].

Positions of Responsibility

Google Developer Groups

Machine Learning Coordinator

Aug 2025 - Present

- Conducted a 12-week ML bootcamp for 200+ students, covering SGD, CNNs, and Kaggle workflows, resulting in 15+ student projects.

DataWorks Club

Computer Vision Coordinator

Aug 2025 - Present

- Mentored 100+ members through OpenCV and PyTorch workshops, building real-time CV pipelines with YOLOv8 and TensorRT targets of 30+ FPS.